

professor, Andrew Lo, who studies investment strategies and is an expert on HFs (he runs one himself), studied the quant meltdown. In Khandani and Lo (2007), they called it “the perfect financial storm.”

Losses at some of the largest, and historically best-performing, quant funds reached 30%,<sup>2</sup> as they did at the Global Equity Opportunities Fund and Global Alpha fund managed by Goldman Sachs Asset Management. (Full disclosure: I was consulting for Morgan Stanley during August 2007 at a quant group that lost \$500 million between the last week of July and August 9.<sup>3</sup>)

Losses of this magnitude in the absence of direct market forces or significant news were shocking to quant HF managers. Their funds were designed to have low volatility. Losses of this size just weren’t supposed to happen. David Viniar, CFO of Goldman Sachs, said:<sup>4</sup> “We were seeing things that were 25-standard deviation moves, several days in a row. There have been issues in some of the other quantitative spaces. But nothing like what we saw last week.”

Matthew Rothman, a quant analyst at Lehman Brothers, commented:<sup>5</sup> “Events that models only predicted would happen once in 10,000 years happened every day for three days.”

It wasn’t just the losses that were shocking. Quants combined several investment strategies that were supposed to have low correlations with each other. Yet the resulting investments plummeted together. Manolis Liodakis, a strategist at Citigroup, exclaimed that “nothing seems to be working. Previously uncorrelated factors have recently been falling with the same pace, leaving investors with very few places to hide.”<sup>6</sup>

Then, as suddenly as the losses arrived, quant returns bounced back on August 10, 2007. The storm left plenty of casualties. Many funds responded to the losses by cutting leverage. But reducing leverage meant that their exposures were much smaller when quant strategies rallied, so their gains did not make up for their losses.

Goldman’s Global Equity Opportunities Fund received an injection of \$3 billion to stabilize the fund, of which \$2 billion came from Goldman and \$1 billion from investors, including Maurice “Hank” Greenberg, the former chairman of AIG (which had to be rescued during the financial crisis a year later), and billionaire Eli Broad. News of the Goldman rescue was announced on August 13, but rumors of it started spreading on August 10 and may even have triggered the rebound. The quant meltdown was the beginning of the end for the Global Equity Opportunities

<sup>2</sup> “The Quant Meltdown: August 2007,” CaseWorks ID #080317 lists August 2007 performance for several quant HFs in Table 2, along with media sources.

<sup>3</sup> See Patterson, S. and A. Raghavan, “August Ambush: How Market Turmoil Waylaid the ‘Quants,’” *Wall Street Journal*, Sept. 7, 2007.

<sup>4</sup> Larsen, P., “Goldman Pays the Price of Being Big,” *Financial Times*, Aug. 13, 2007.

<sup>5</sup> Whitehouse, K., “One Quant Sees Shakeout for the Ages – ‘10,000 years,’” *Wall Street Journal*, Aug. 11, 2007.

<sup>6</sup> Xydias, A., “Market Turmoil is ‘Perfect Storm’ for Quant Funds,” *Bloomberg*, Aug. 10, 2007.